SYSTEMIC COGNITION

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SYSTEMIC COGNITION IS A CONSTRUCTIVIST FRAMEWORK

Combining humanities and social sciences with systems science to support HSI in complex socio-technical systems (STSs).

Providing ontological support for discussing this sociotechnical evolution and to develop HSI

- not only based on a Human-Centered Design (HCD) approach,
- but also by focusing on society,
- which is becoming increasingly immersed in a world equipped with artificial resources

Cross-fertilizing and articulating logical (procedural) and teleological (declarative) representations

Capturing domain experience from the bottom up and constructing STSs incrementally from the top down

Introducing PRODEC, a dual procedural-declarative method and platform

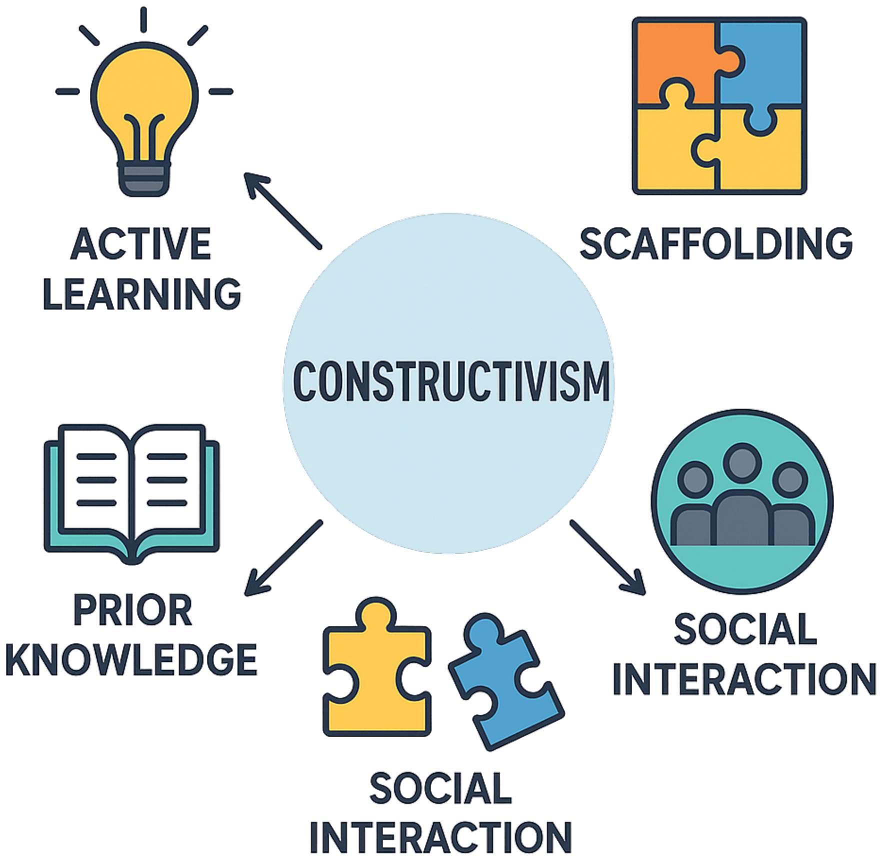
Illustrating using 4 concrete real-world applications

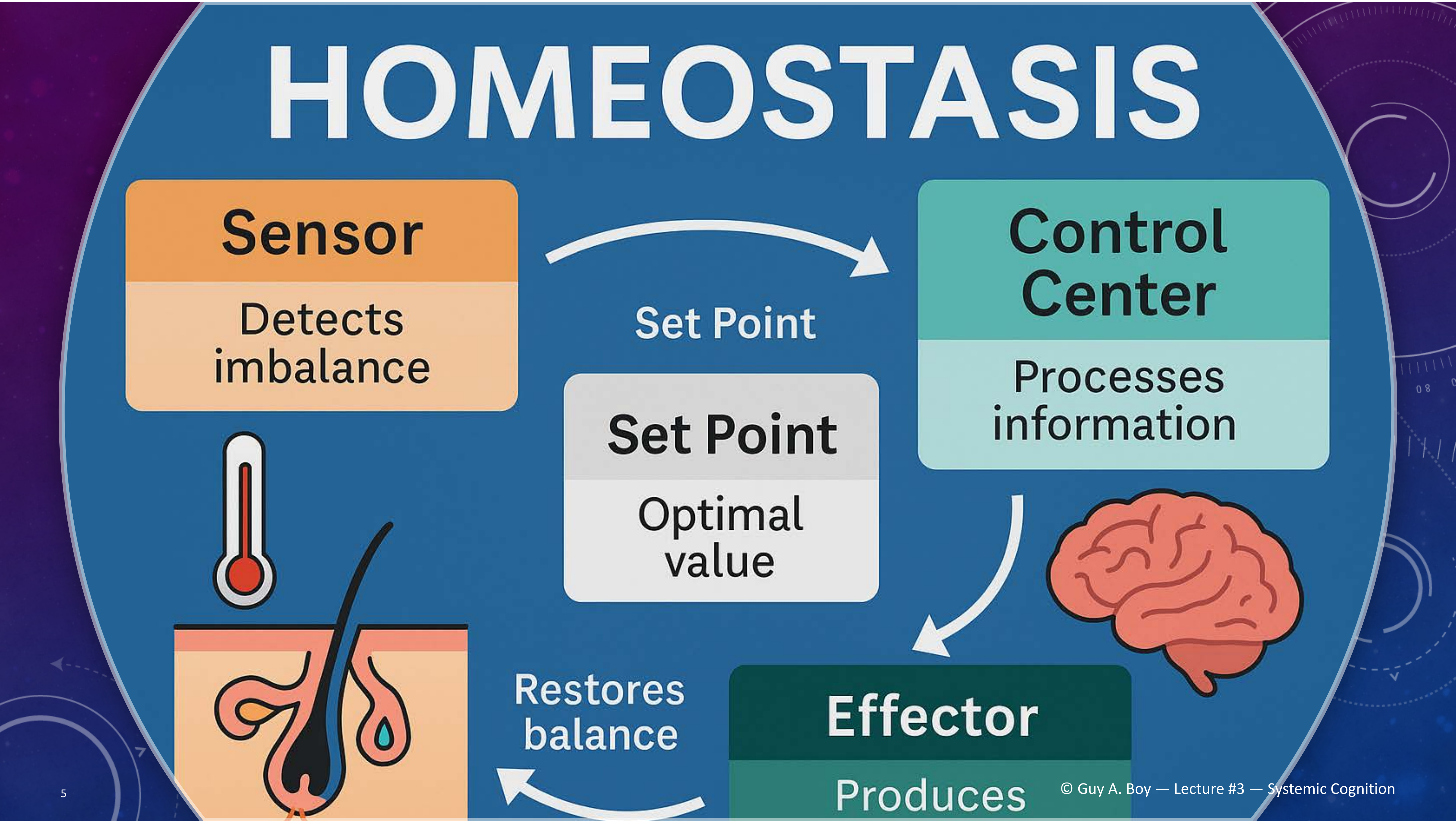
https://www.mdpi.com/2079-8954/14/1/15

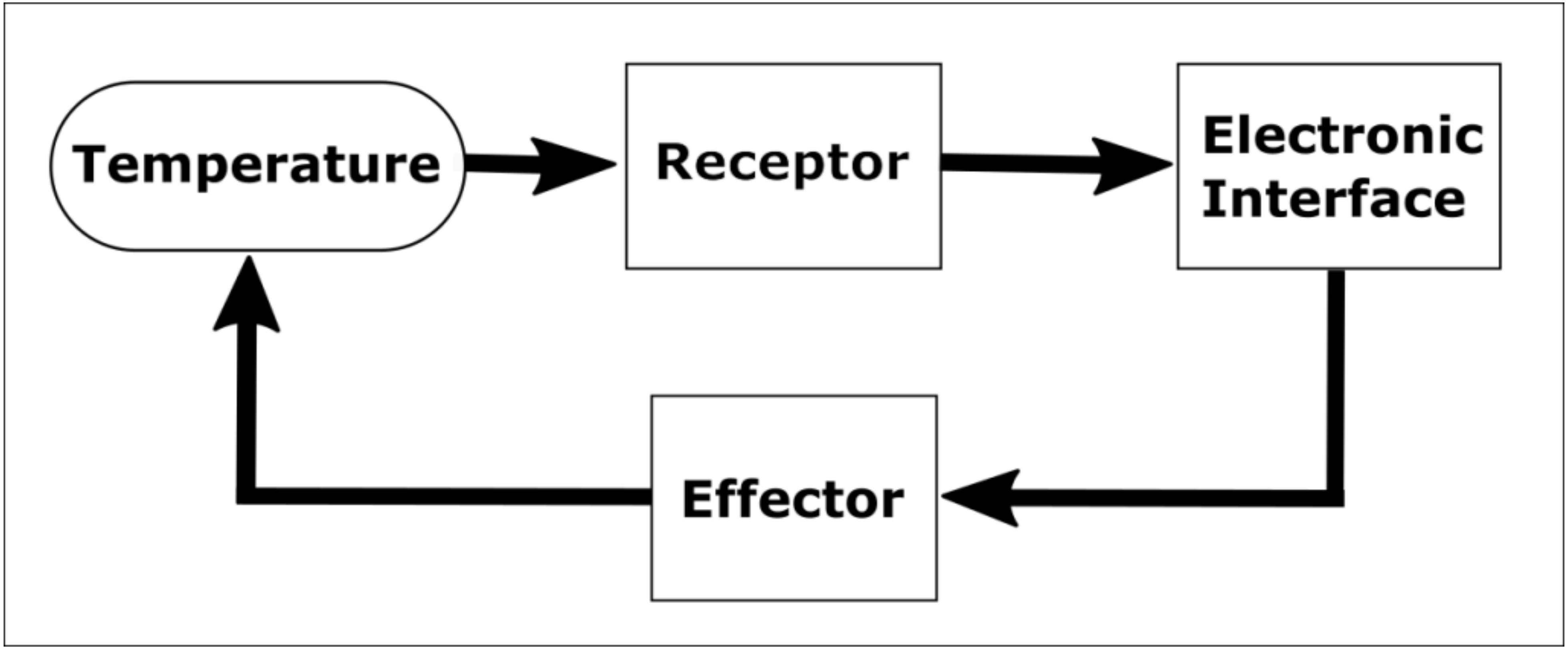
WHAT CONSTRUCTIVISM IS ABOUT…

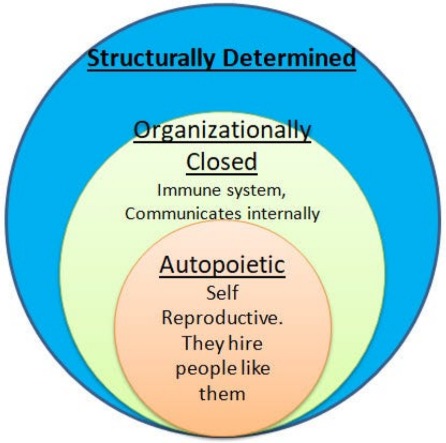
Constructivism is a theory
that suggests that learners
do not passively acquire knowledge
through direct instruction.

Instead, they *construct* their understanding
through experiences and social interaction,
integrating new information
with their existing knowledge.


ACTIVE LEARNING
SCAFFOLDING
CONSTRUCTIVISM
PRIOR KNOWLEDGE
SOCIAL INTERACTION
SOCIAL INTERACTION


HOMEOSTASIS
Sensor
Detects imbalance
Set Point
Control Center
Processes information
Set Point
Optimal value
Restores balance
Effector
Produces


Temperature
Receptor
Electronic Interface
Effector



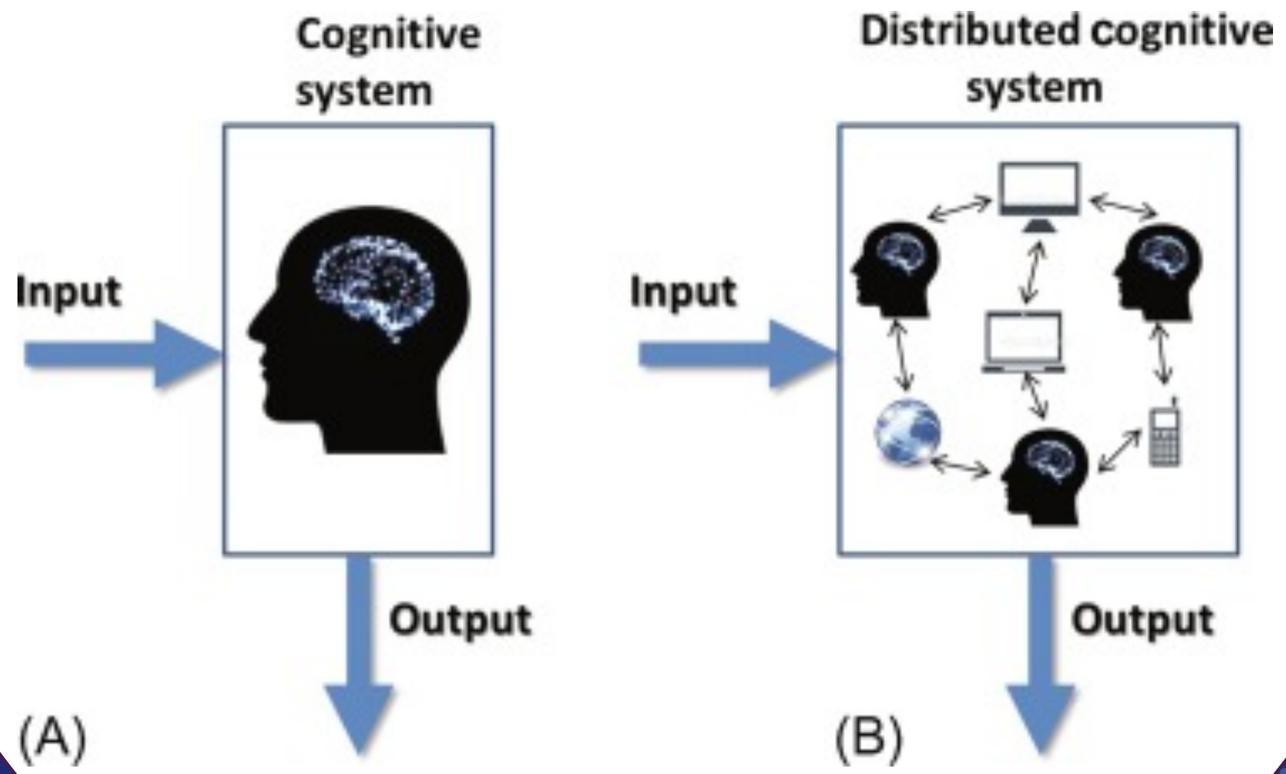
Autopoietic Systems are self-contained,
Self-reproductive, closed systems, and within its
environment it's a dynamic network of interactions

AUTOPOIESIS

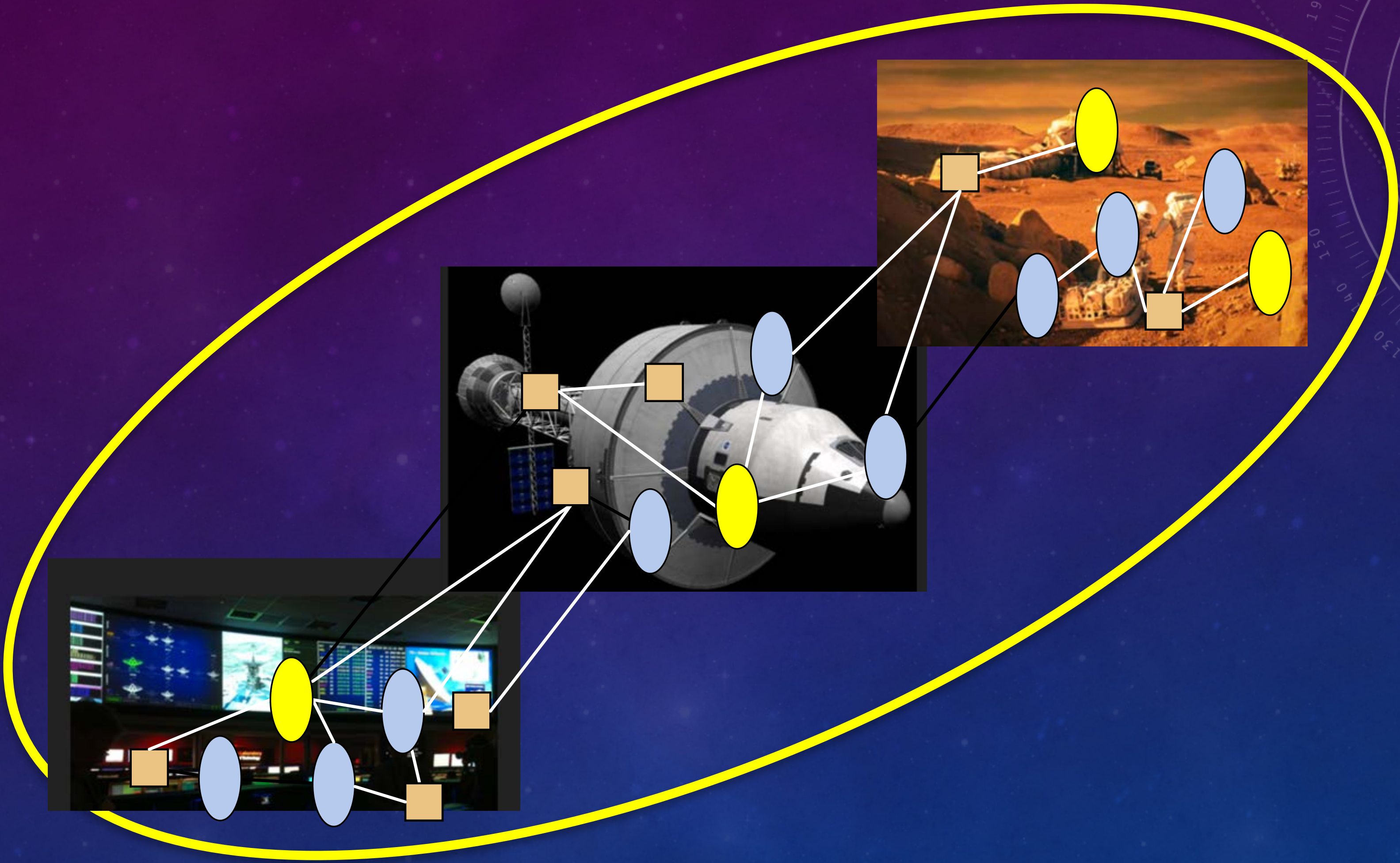
DISTRIBUTED COGNITION

« The emphasis on finding
and describing ‘knowledge structures’
that are somewhere ‘inside’
the individual encourages us
to overlook the fact that human cognition
is always situated
in a complex sociocultural world
and cannot be unaffected by it. »

– Edwin Hutchins, cognitive scientist
and author of *Cognition in the Wild (1995)*



DEALING WITH COMPLEXITY

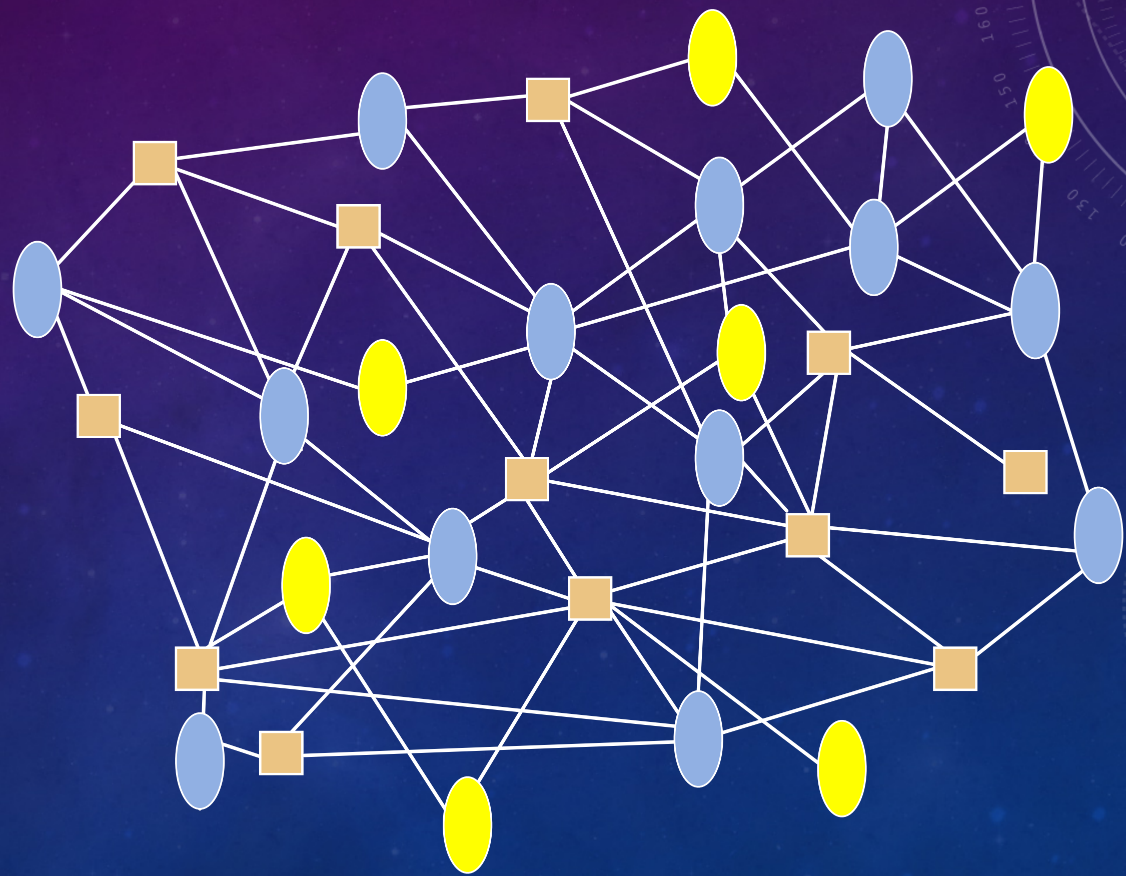


A COMPLEX SYSTEM AS A LIVING ORGANISM

Separability
a crucial issue

Complexity
in the connexions
as well as
in the agents themselves

Emergents Functions
&
the **maturity** issue



PROPERTIES OF A COMPLEX SYSTEM

- a large **number** of components and interconnections among them
- many people involved in its **life cycle**

  (design, development, manufacturing, operations, maintenance and dismantling)
- **emergent** properties and behaviors not included in the components
- complex adaptive mechanisms and behaviors (**adaptability**)
- nonlinearities and possible chaos (**unpredictability**)

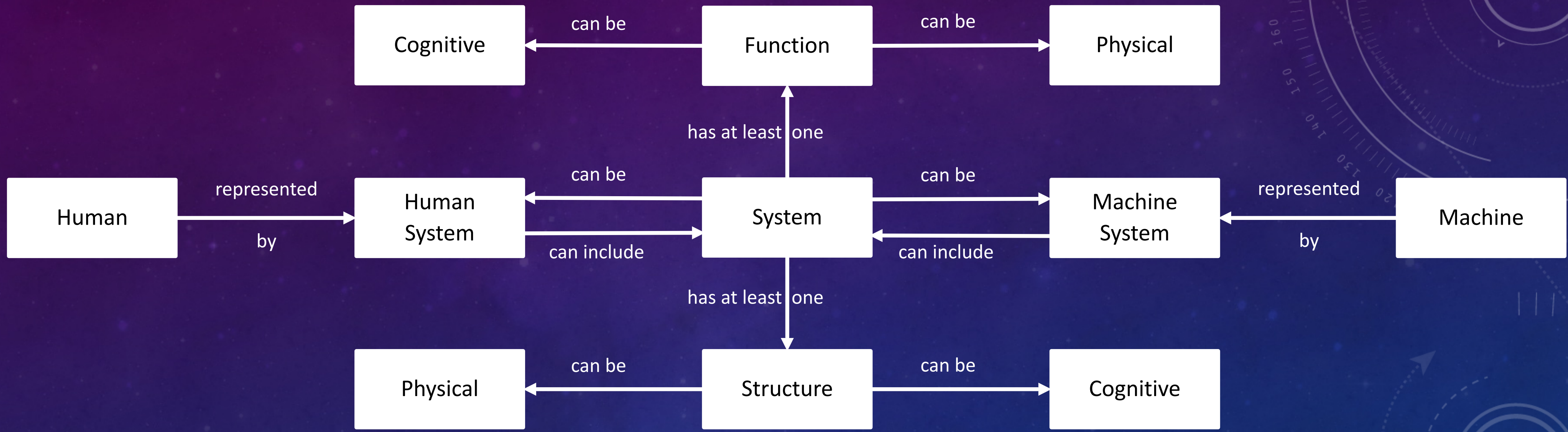
METHODS & TOOLS

- Design thinking: creativity & ideation
- Task analysis & Scenario-based design
- Conceptual modeling
- **PRODEC**
- Rapid prototyping
- Human-in-the-loop simulation
- Evaluation methods and tools
- Agile development
- Organizational design & management

INTEGRATION

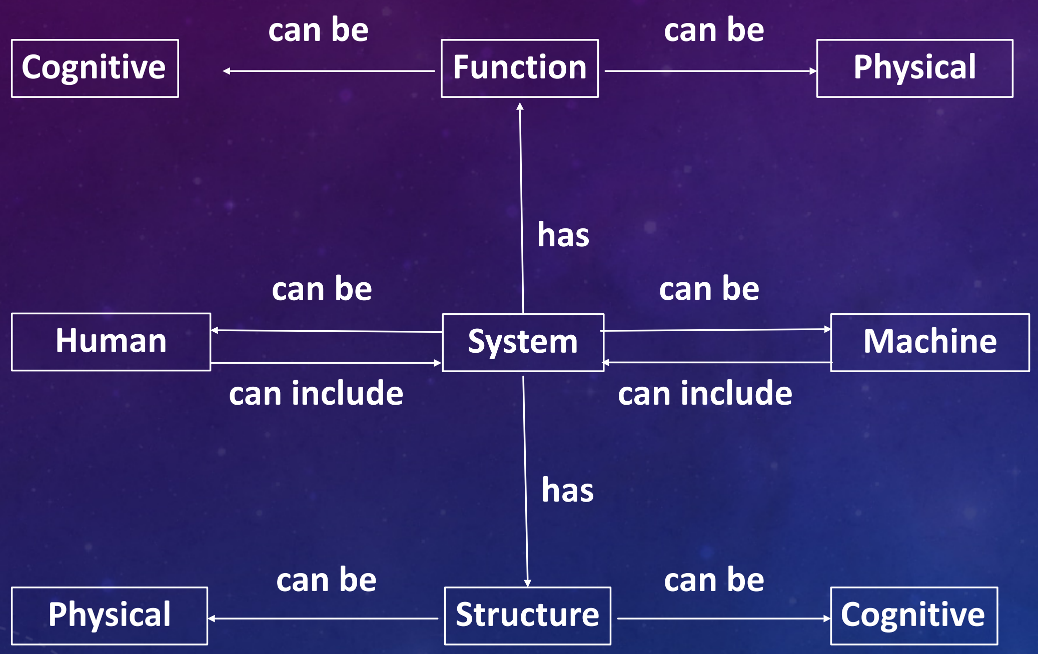
INTEGRATION

THE NEED FOR A SYSTEMIC ONTOLOGY

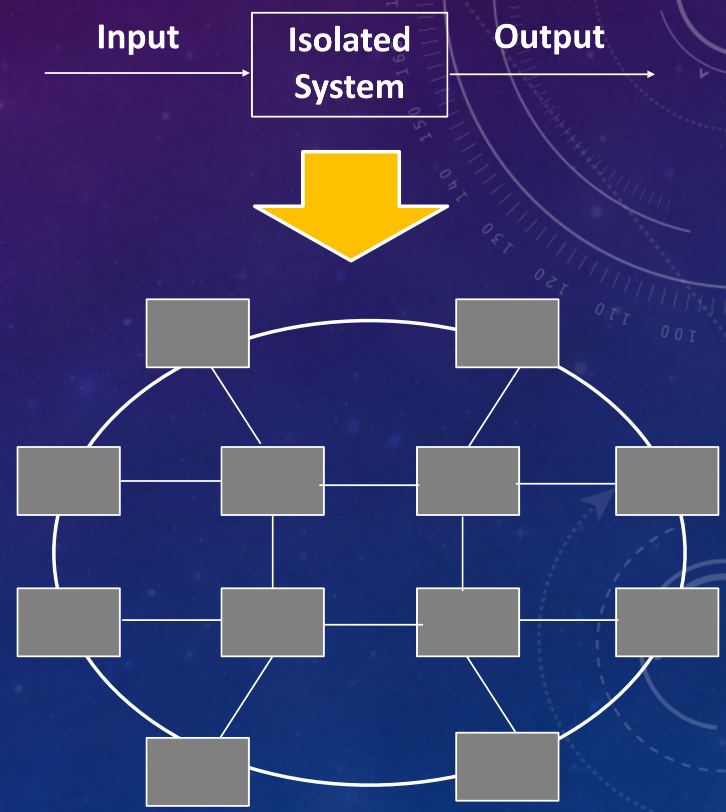


SYSTEMS ARE REPRESENTATIONS OF NATURAL OR ARTIFICIAL ENTITIES

WHAT IS A SYSTEM?

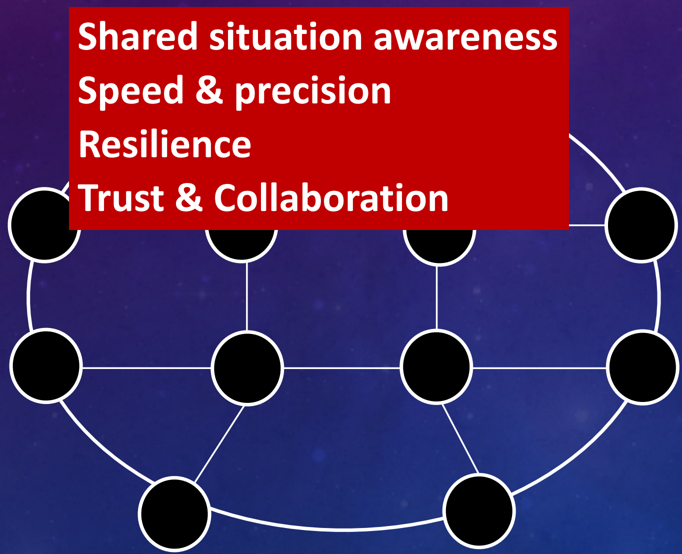


Systems include Humans and Machines…



Interconnected System of Systems

SYSTEM = FUNCTION + STRUCTURE

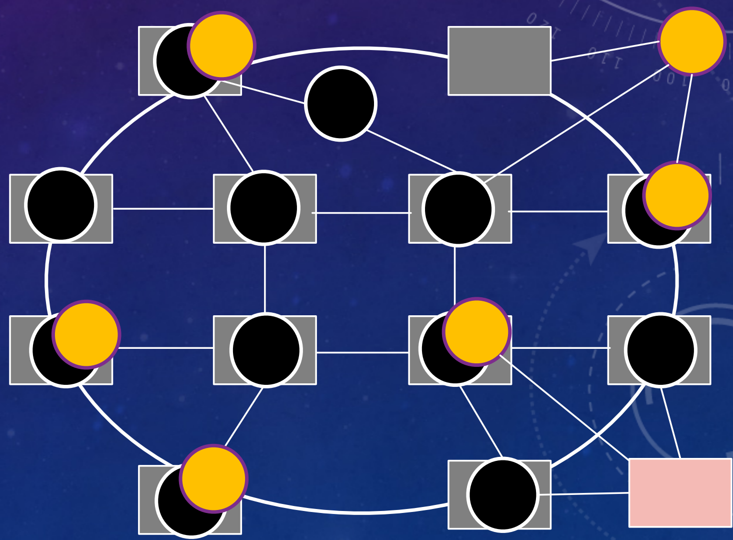


Interconnected Functions of Functions

Emergent Structures

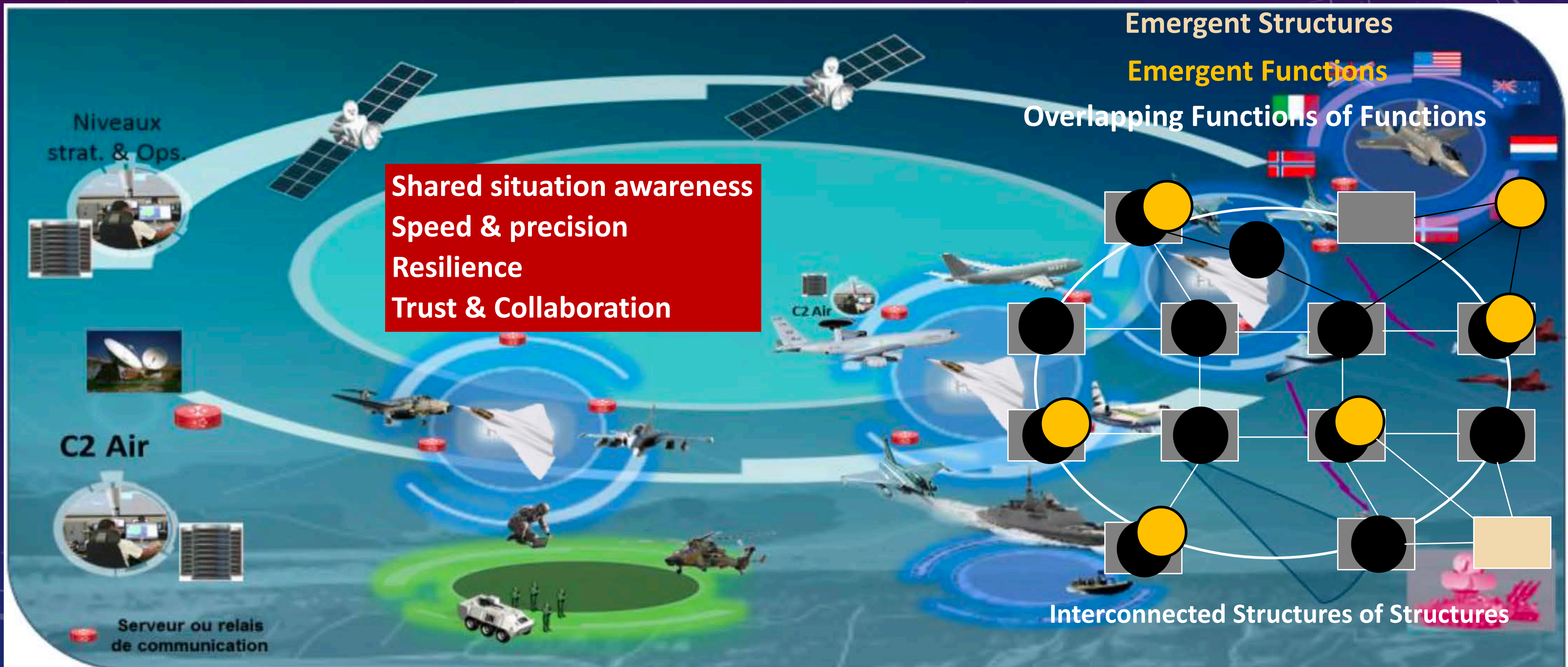
Emergent Functions

Overlapping Functions of Functions

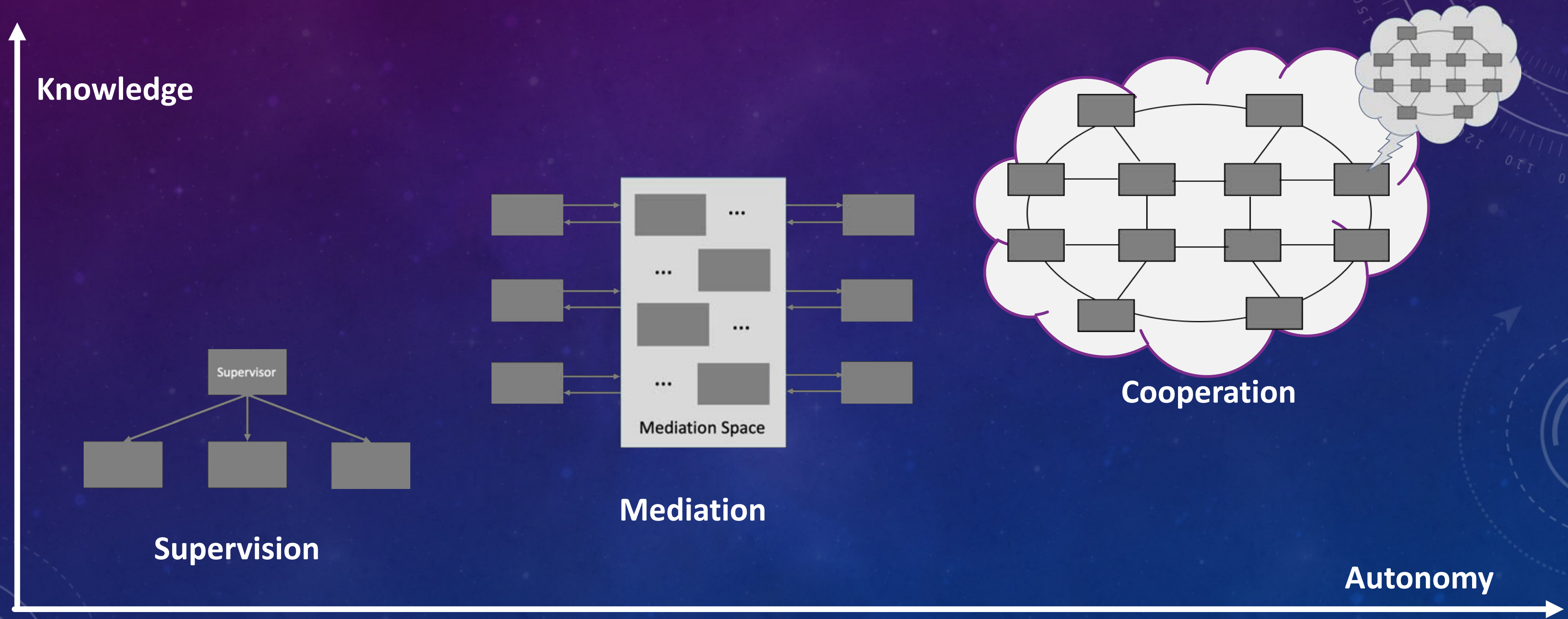


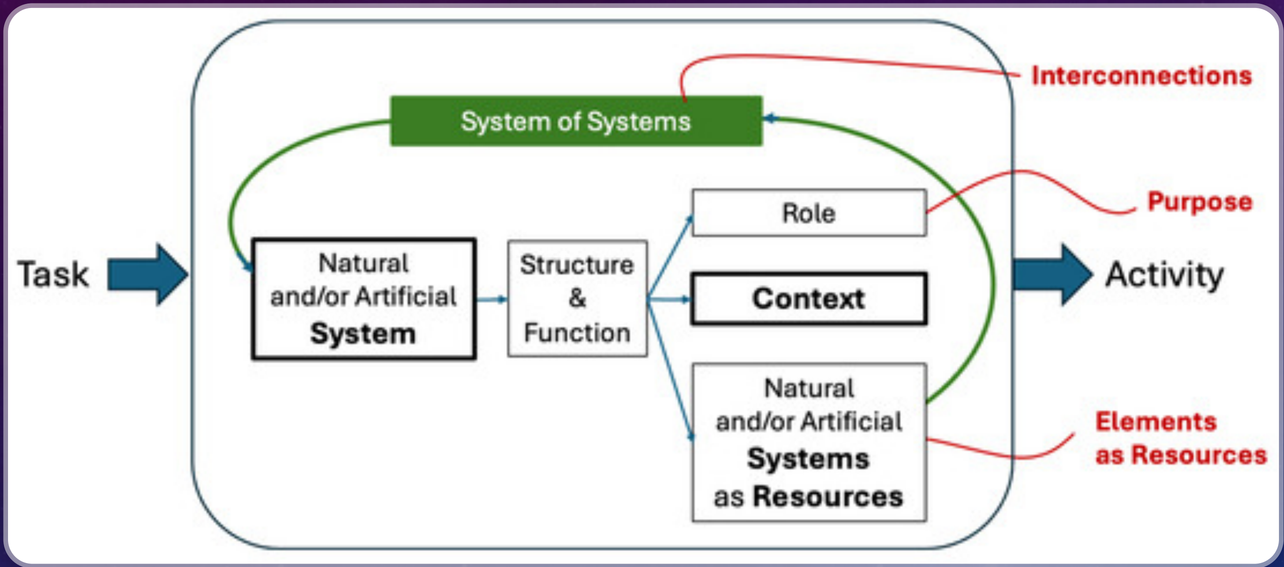
Interconnected Structures of Structures

SYSTEM = FUNCTION + STRUCTURE



SYSTEMIC INTERACTION MODELS…
… AND AUTHORITY SHARING

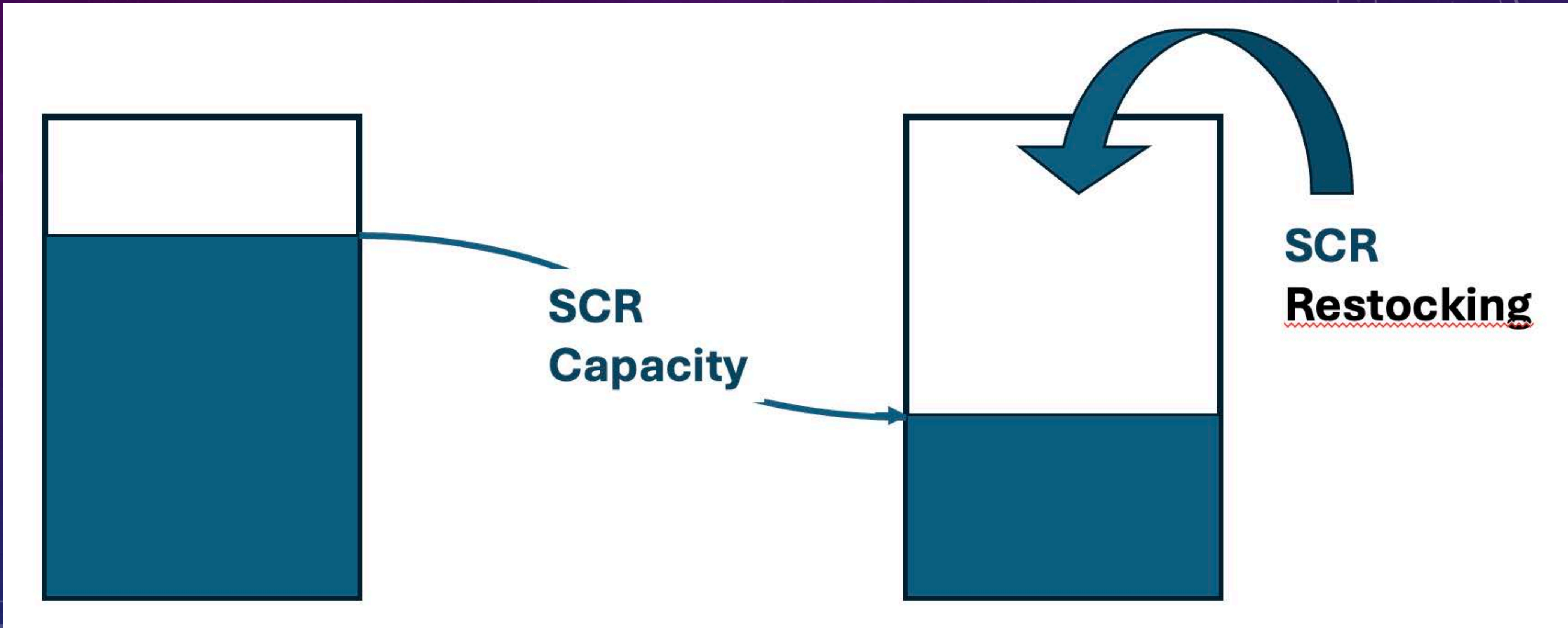


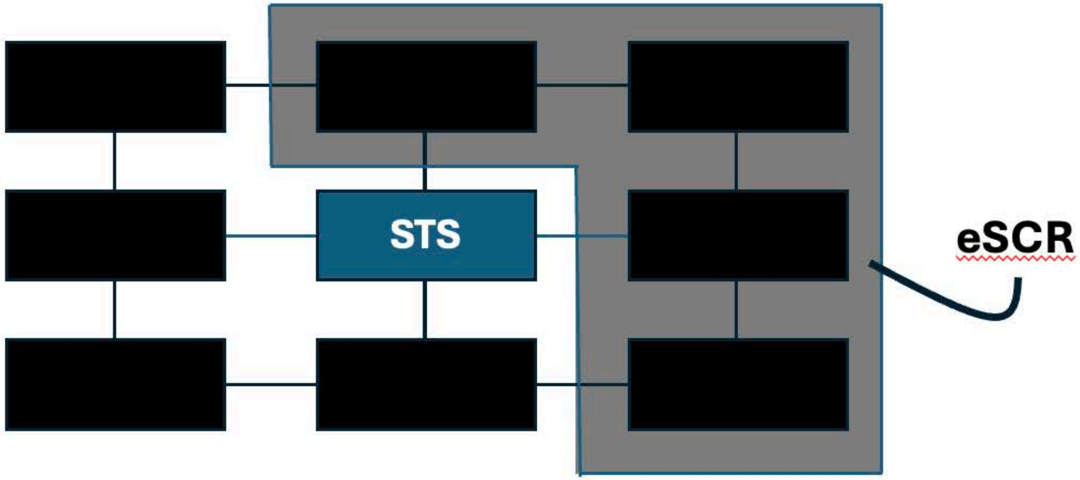


THE LOGICAL AND TELEOLOGICAL DEFINITION OF A SOCIOTECHNICAL SYSTEM

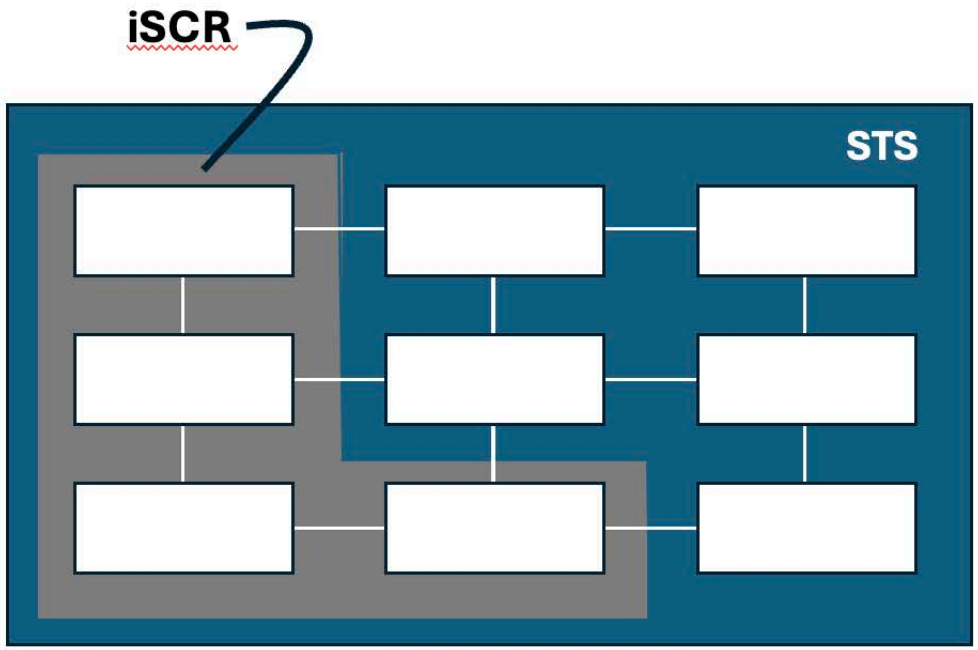
- The logical part
  - Prescribed task
  - Effective activity
- The teleological part
  - Structure & function
  - Role (goal-driven)
  - Context (event-driven)
  - Resources (systems)

SYSTEMIC COGNITION RESOURCE (SCR)





External Support

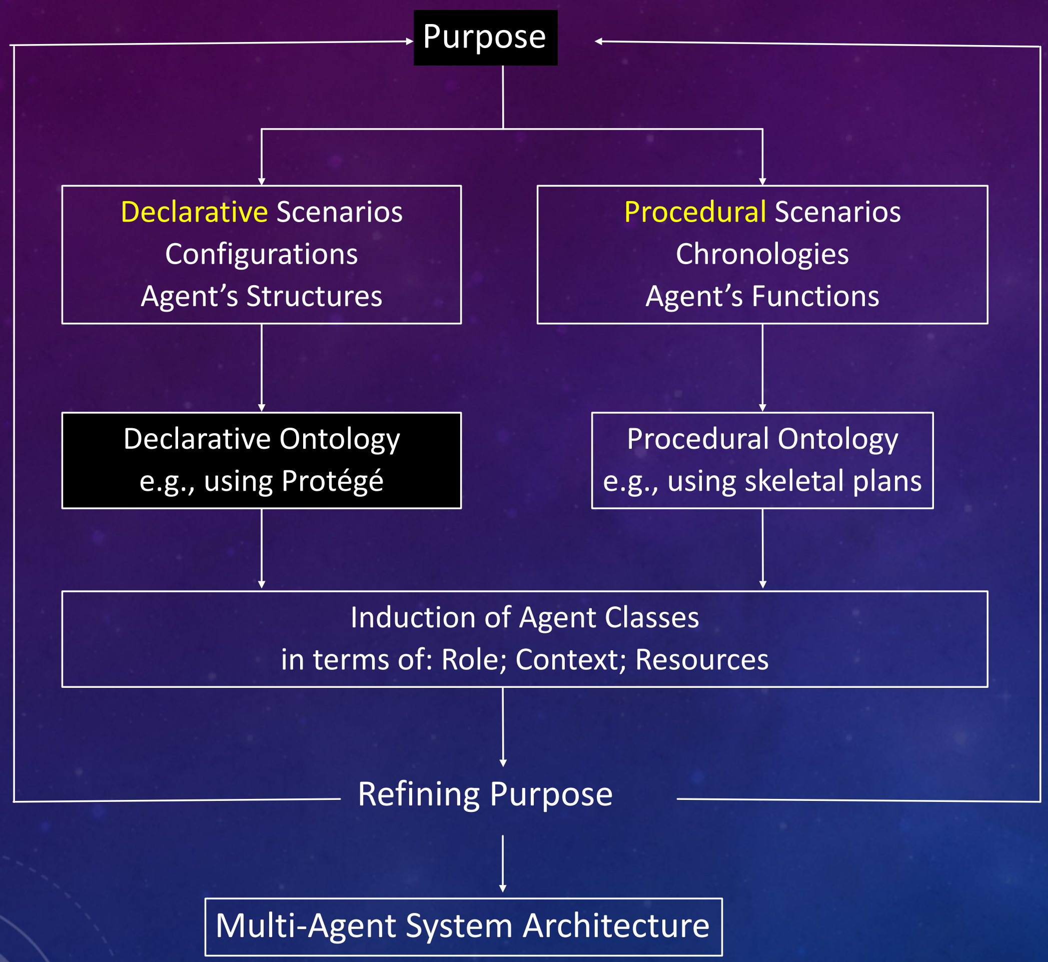


Competencies

INTEGRATION

FROM PURPOSE TO MEANS

FROM PURPOSE TO MEANS



What do we want to do?

Analysis of the existing so far…

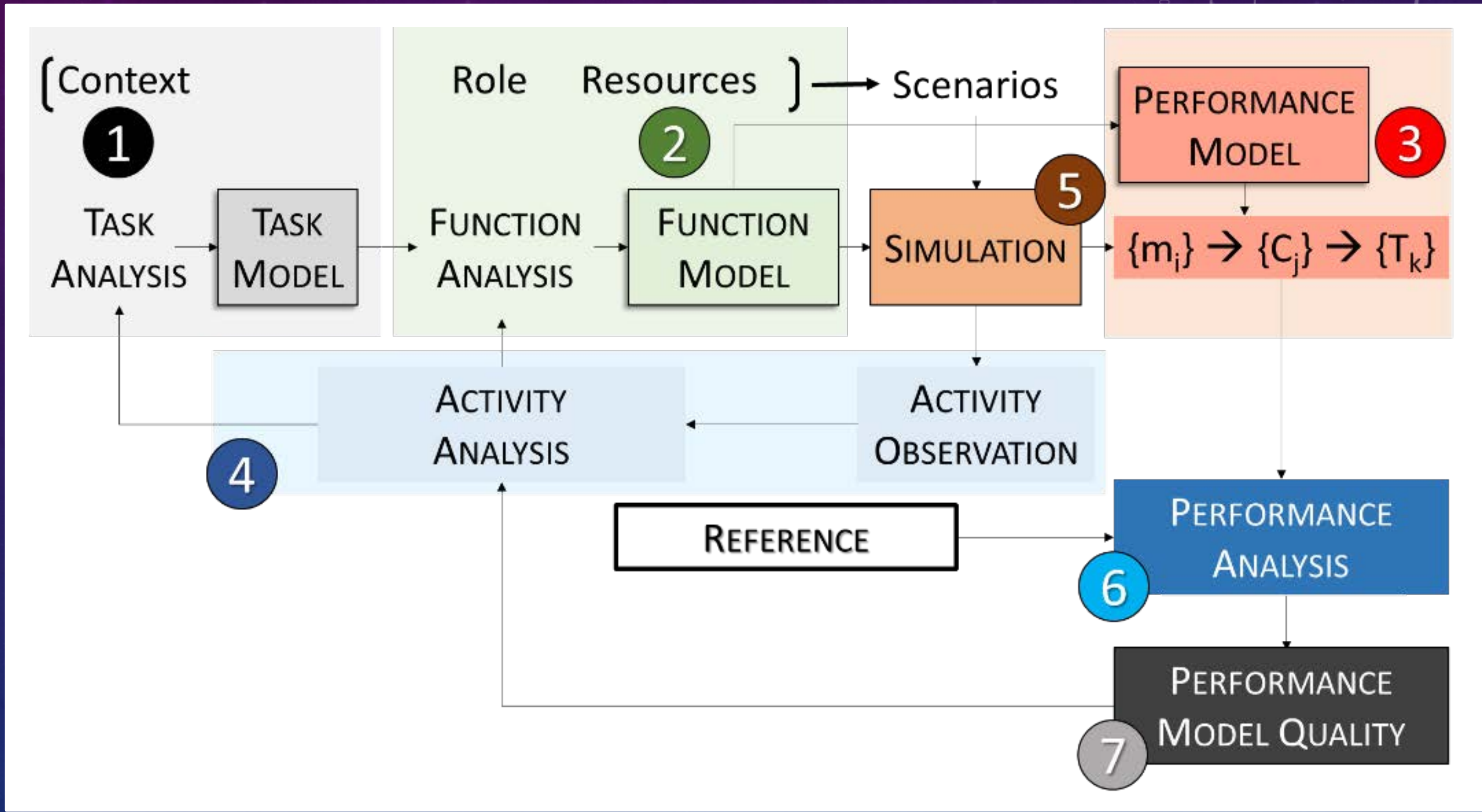
Anticipating possible futures…

Making a Multi-Agent Ontology

Becoming more generic…

PRODEC

MOHICAN PRODEC



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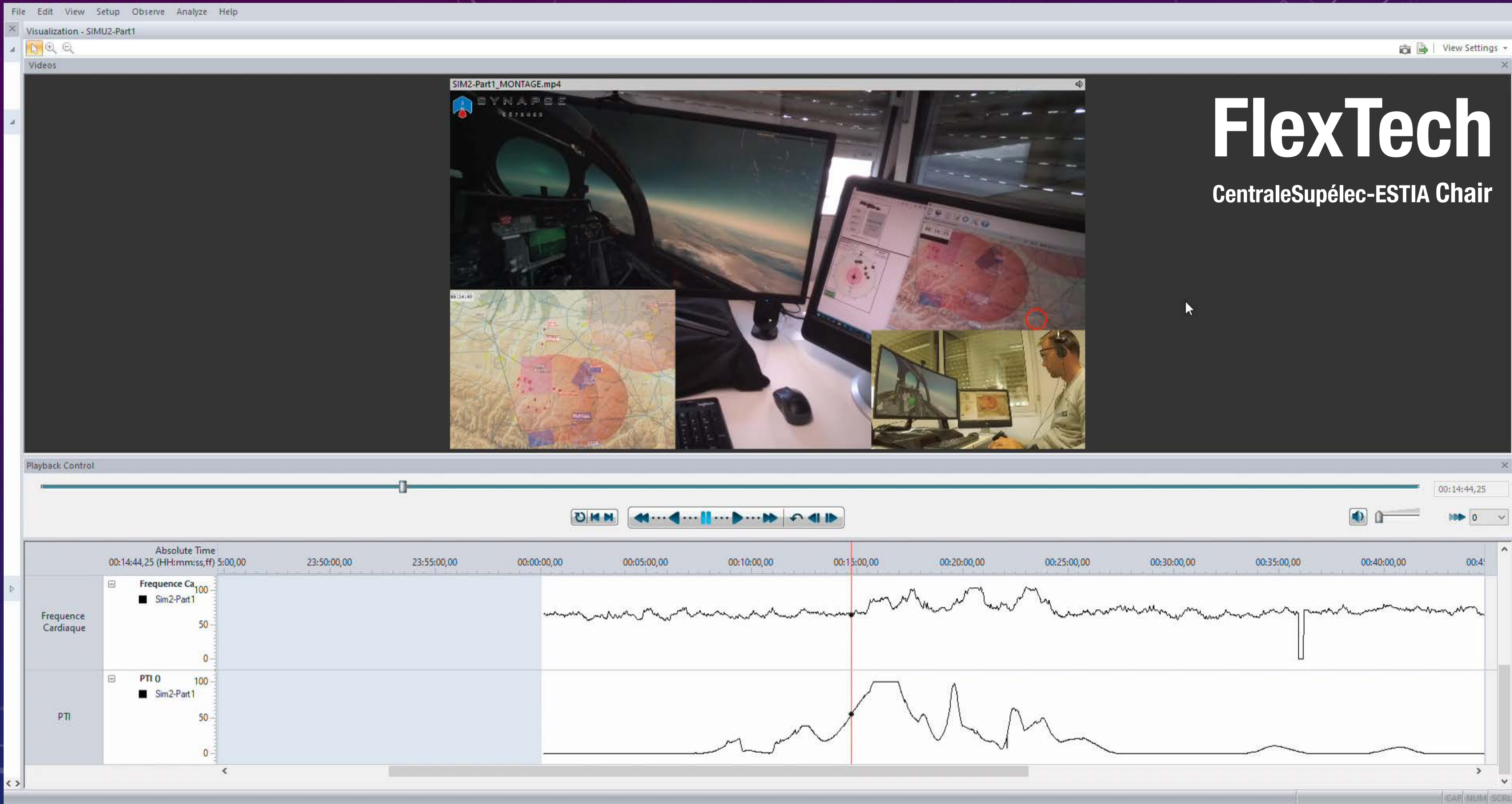
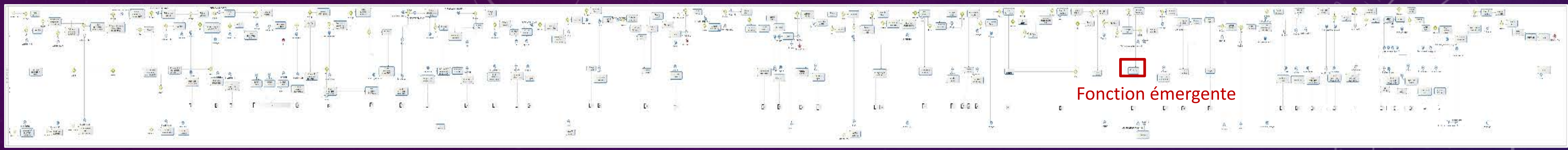
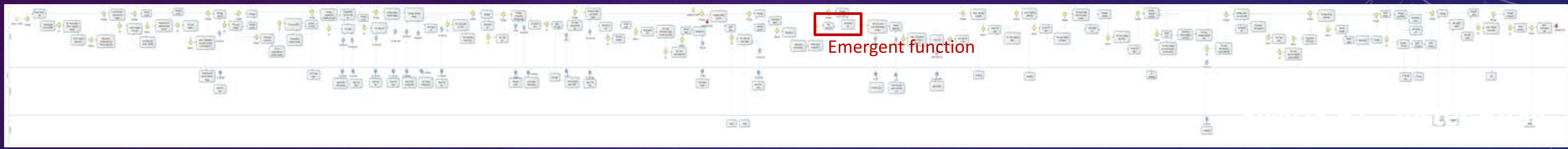
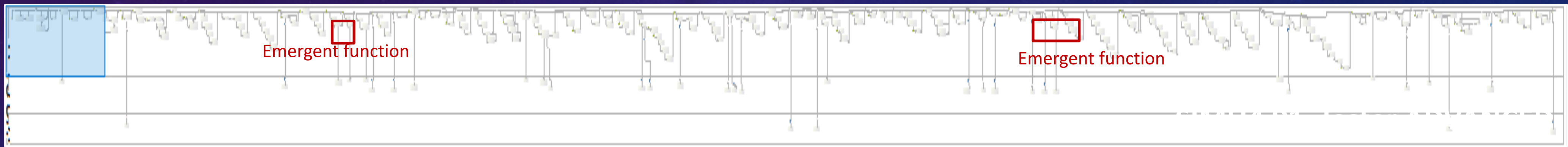
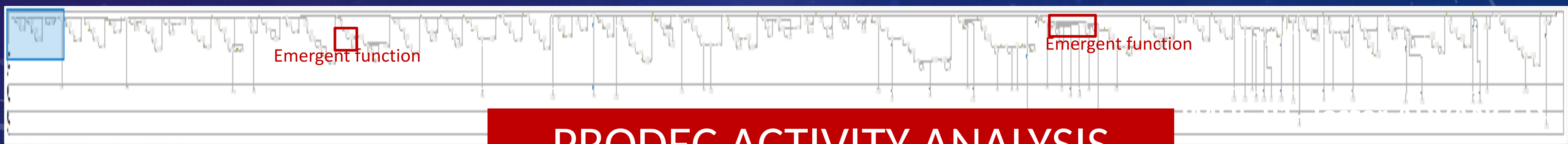










PRODEC ACTIVITY ANALYSIS

PRODEC

PRODEC

OPERATIONS

PROCEDURAL SCENARIOS

CONTEXTUAL ARCHITECTURE …

… SYSTEMIC ARCHITECTURE

DECLARATIVE CONFIGURATIONS

DESIGN ENGINEERING

INTEGRATION OF PRODEC INTO INGESCAPE

SINGLE PILOT OPERATIONS (SPO)?

- Continuity vs. discontinuity: from 5 to 4, from 4 to 3, from 3 to 2, …
- Today, from 2 to 1?
- Digitalization → job change
- Crew number reduction on aircraft or definition of a new job for handling a drone?

From CFA to PRODEC

HSI	HUMAN SYSTEMS INTEGRATION
=	
HCD	HUMAN-CENTERED DESIGN
+	
SE	SYSTEMS ENGINEERING

From Rigid Automation to Flexible Autonomy

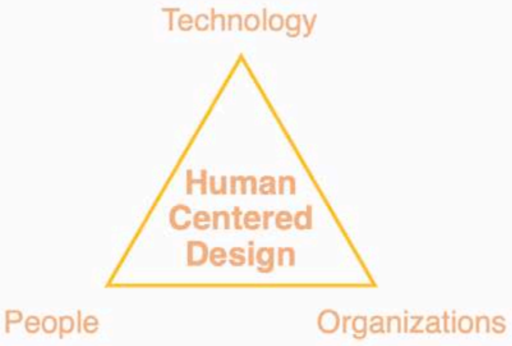
FlexTech Chair philosophy is based on **human-centered design** (HCD) and creativity, collaborative work, cognitive engineering, complexity analysis, organization design and management, modeling and human-in-the-loop simulation, artificial intelligenece, advanced interaction media, and the study of life-critical systems.

HCD combined with Systems Engineering leads to Human Systems Integration (HSI).

**We automated a lot during the 20th century increasing safety, efficiency and comfort in nominal situations,
but leading to rigidity in off-nominal situations.**

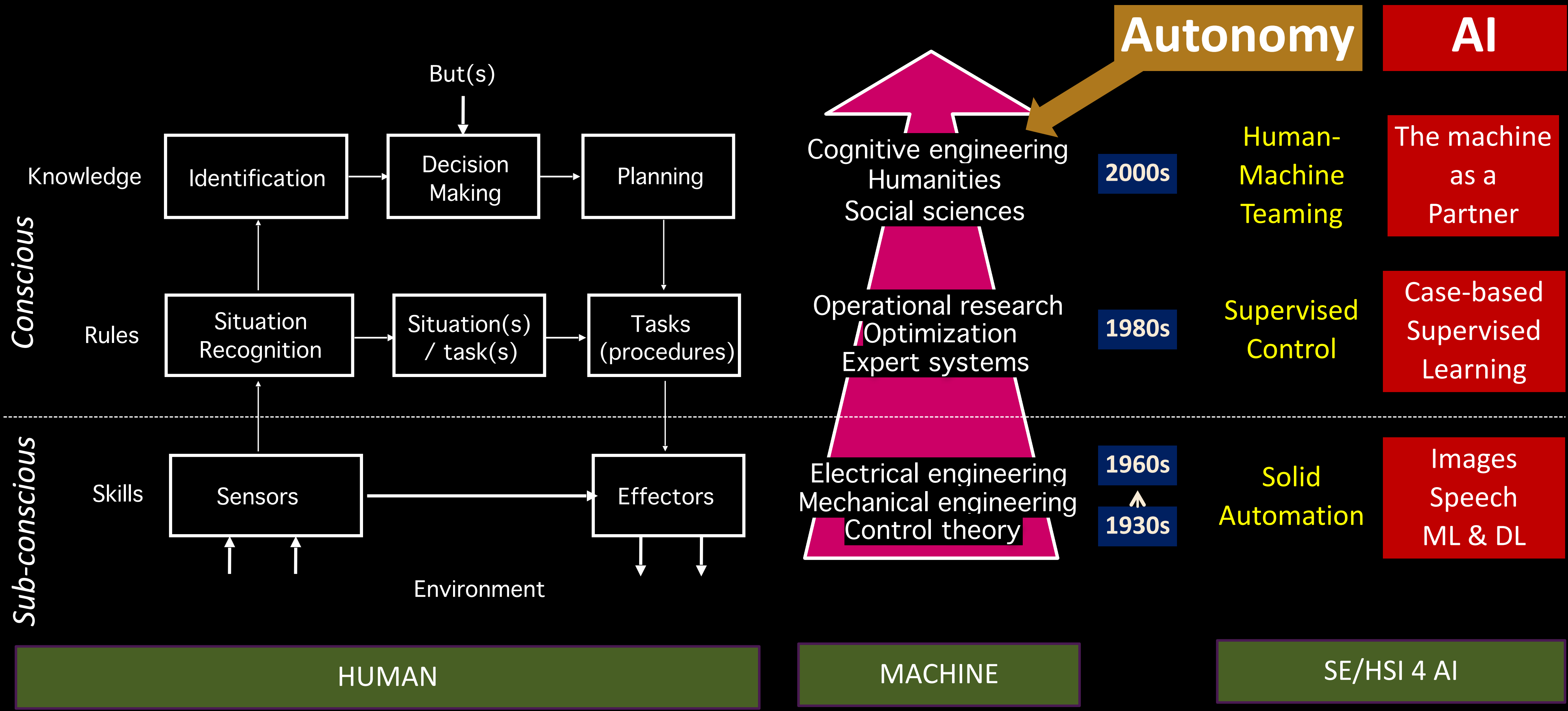
It is time to develop research and innovation on flexibility that increases autonomy of technology, organizations and people.

This is the shift from HighTech to FlexTech.



https://www.flextechchair.org/about.html

WHAT DO WE MEAN BY ARTIFICIAL INTELLIGENCE?



TECHNOLOGY-CENTERED ENGINEERING & MECHANICAL INDUSTRY

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20TH CENTURY APPROACH

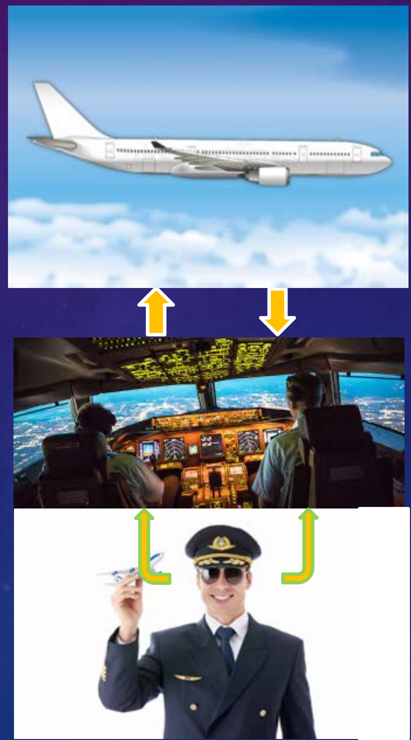
Engineering
Ergonomics
HCI
Automation

From hardware
… to software

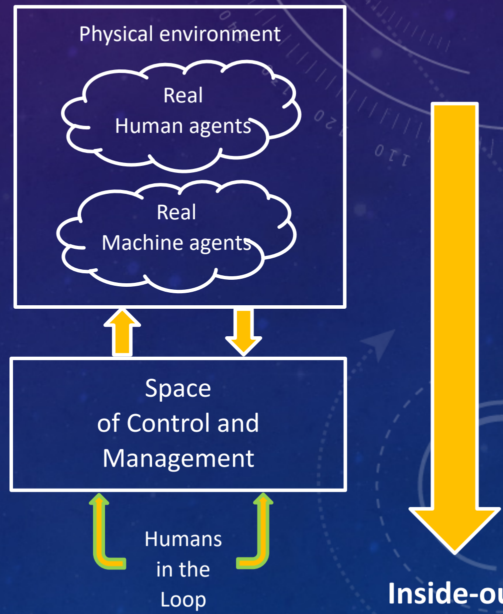
Engineering

Ergonomics
Automation

Pilots
+
Aero personnel



People adapt
to Machines



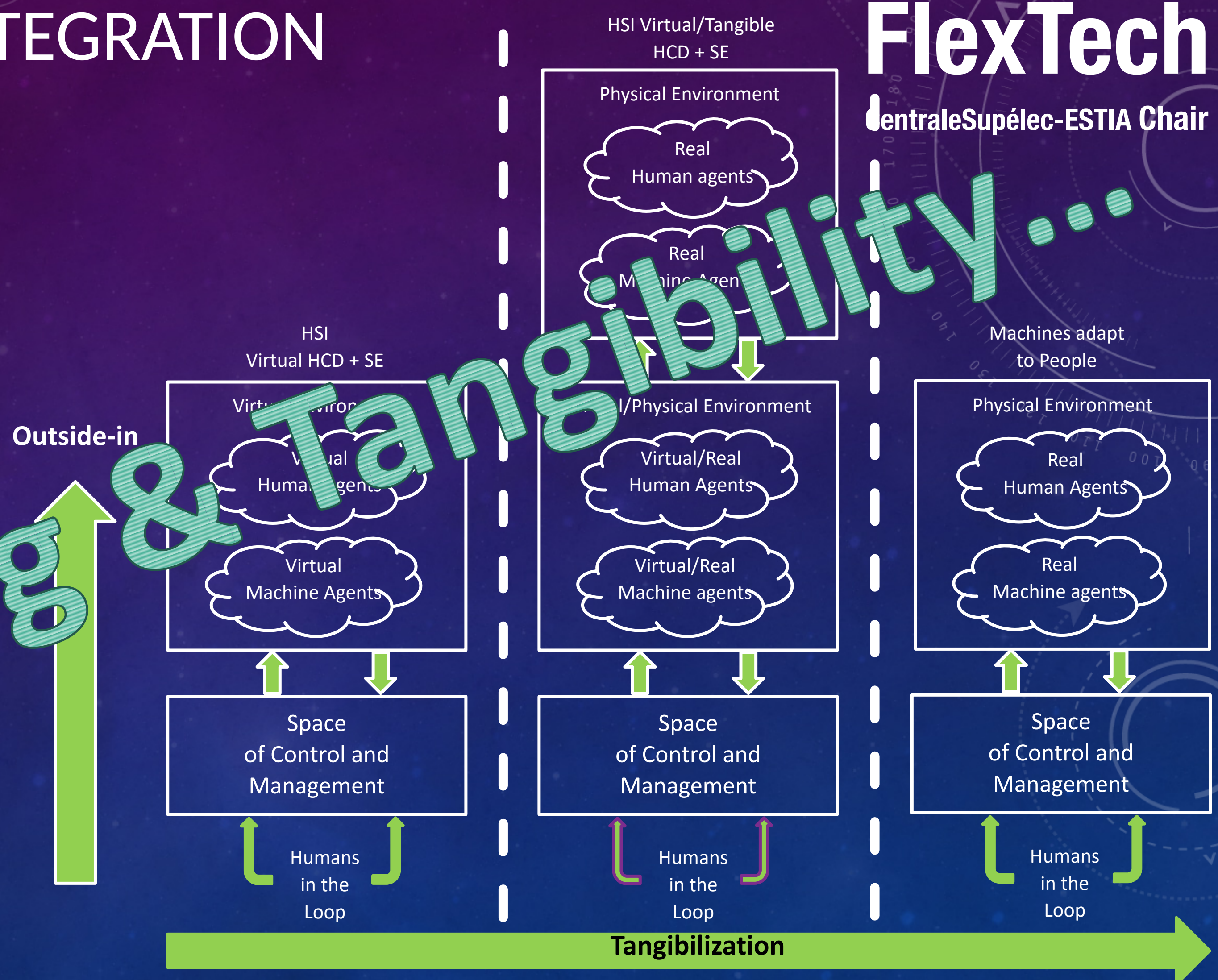
HUMAN SYSTEMS INTEGRATION & DIGITAL INDUSTRY

21ST CENTURY APPROACH

Human-centered design (HCD)
Systems engineering (SE)
Human-in-the-loop digital simulation
Artificial Intelligence & Data Science
Organization sciences

From software
… to hardware

Meaning & Tangibility…

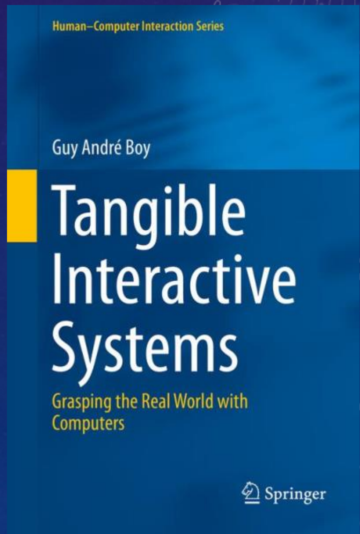


TANGIBILITY: SYSTEMIC ATTRIBUTES

- Complexity: separability, interconnectivity, collaboration, trust, ...
- Maturity: TRLs & HRLs & ORLs
- Flexibility (design & operations): safety modes, reversibility, FlexTech, ...
- Stability/Resilience: passive vs. active, resilience, crisis management, ...
- Sustainability: design rationale, knowledge management, ...

+ Sociotechnical Factors

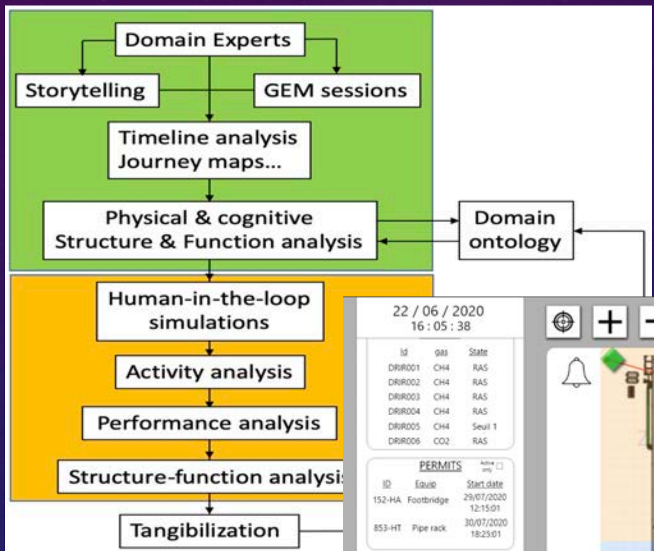
Shared situational awareness
Cooperative decision-making
Harmonized risk-taking
Trust and collaboration

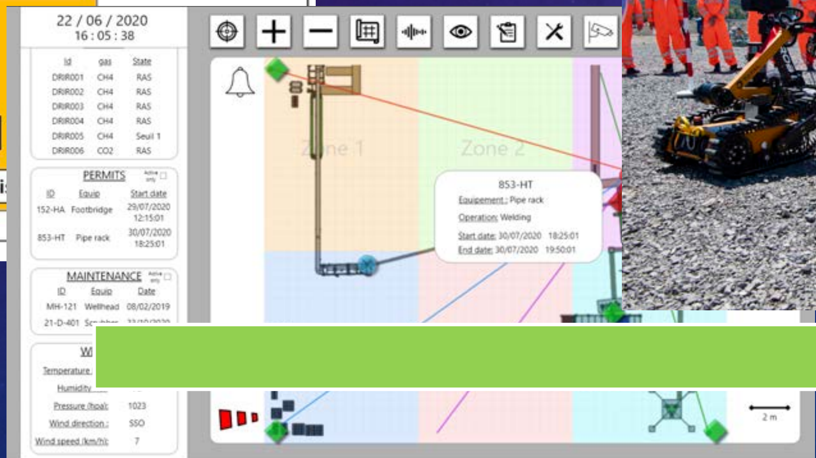


MULTI-AGENT TELEROBOTIC SYSTEMS FOR OFFSHORE OIL & GAS WELLS

Situational awareness, decision-making & risk-taking

Using the PRODEC method and human-in-the-loop simulation





Tangibilization

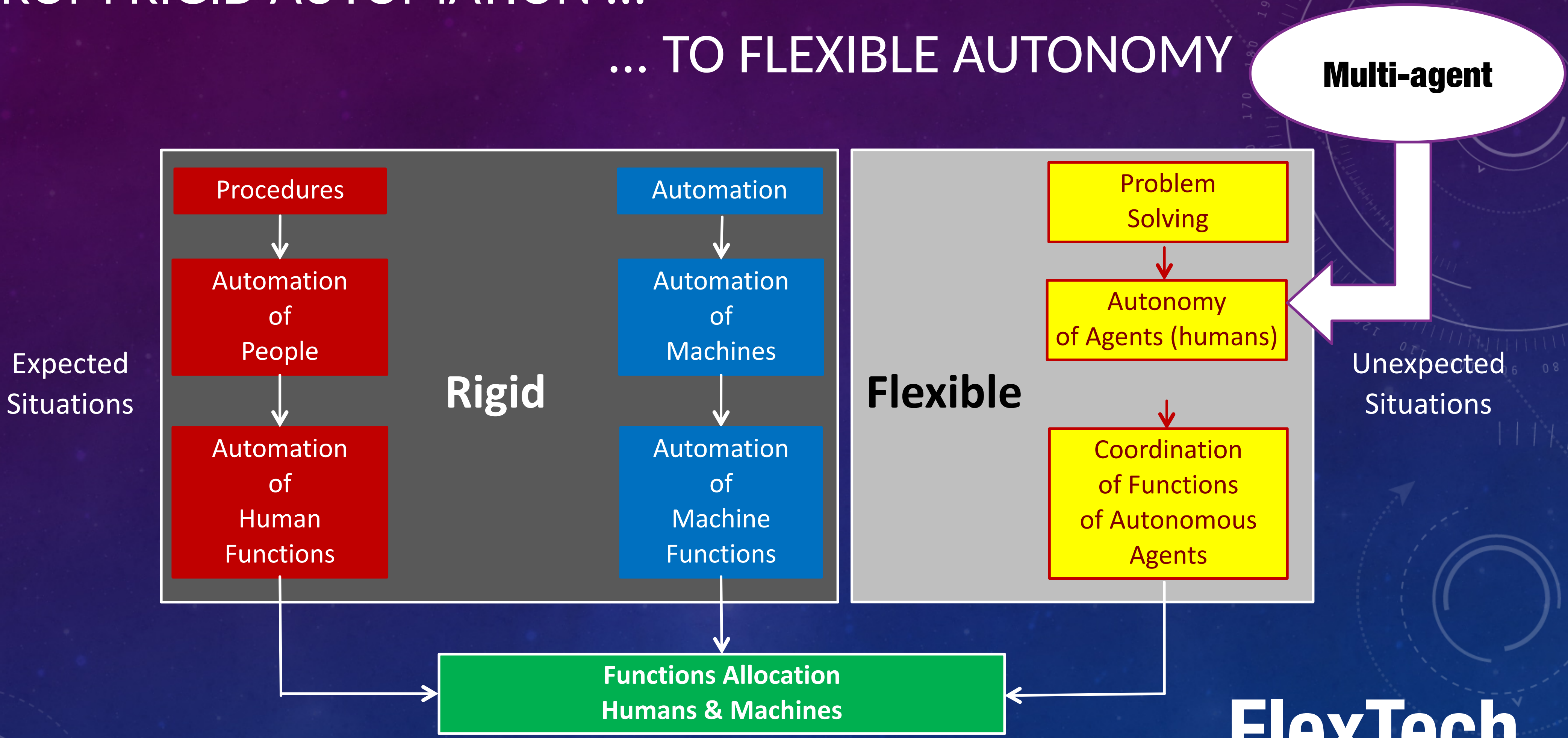
DEALING WITH THE UNEXPECTED…

APOLLO 13 CO_2 …

… COLLABORATIVE PROBLEM-SOLVING

FROM RIGID AUTOMATION …

… TO FLEXIBLE AUTONOMY



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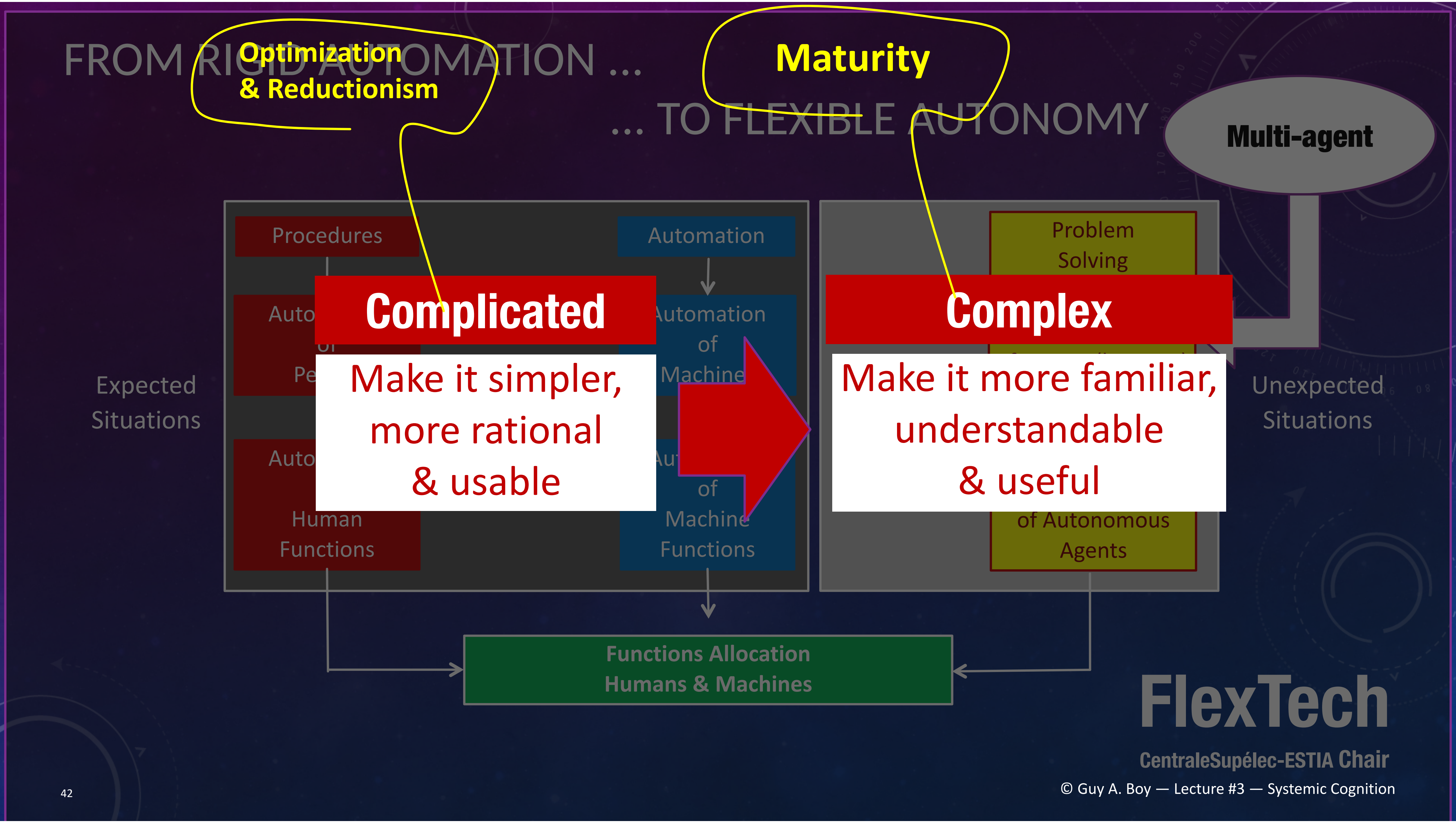
FROM RIGID AUTOMATION …
… TO FLEXIBLE AUTONOMY
Optimization
& Reductionism
Maturity
Multi-agent
Procedures
Automation
Automation
of
Machine
Human
Functions
Machine
Functions
Problem
Solving
of Autonomous
Agents
Complicated
Make it simpler,
more rational
& usable
Complex
Make it more familiar,
understandable
& useful
Expected
Situations
Unexpected
Situations
Functions Allocation
Humans & Machines
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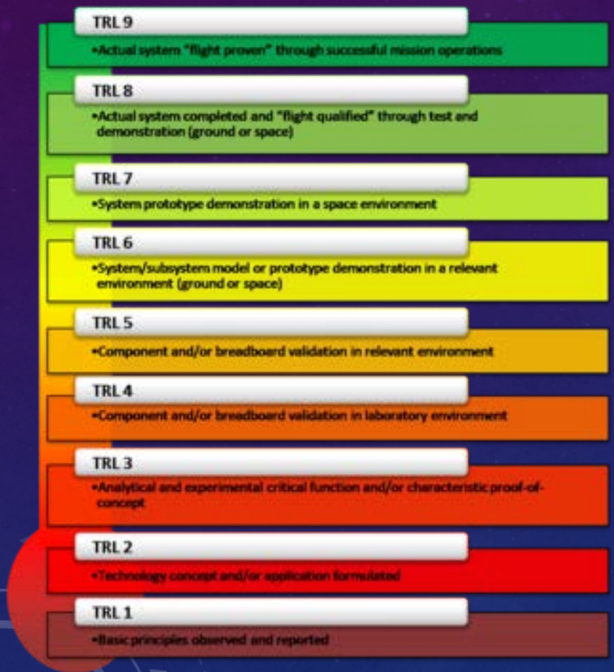
© Guy A. Boy — Lecture #3 — Systemic Cognition

READINESS LEVELS

Technology (TRL)



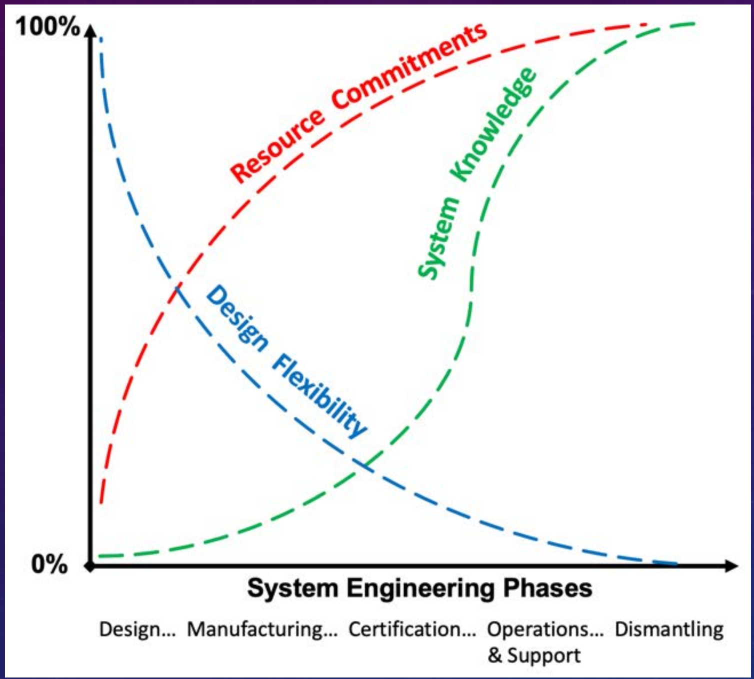
Human (HRL)

HRL	Description
1	Relevant human capabilities, limitations, and basic human performance issues and risks identified
2	Human-focused concept of operations defined and human performance design principles established
3	Analyses of human operational, environmental, functional, cognitive, and physical needs completed, based on proof of concept
4	Modeling, part-task testing, and trade studies of user interface design concepts completed
5	User evaluation of prototypes in mission-relevant simulations completed to inform design
6	Human-system interfaces fully matured as influenced by human performance analyses, metrics, prototyping, and high-fidelity simulations
7	Human-system interfaces fully tested and verified in operational environment with system hardware and software and representative users
8	Total human-system performance fully tested, validated, and approved in mission operations, using completed system hardware and software and representative users
9	System successfully used in operations across the operational envelope with systematic monitoring of human-system performance

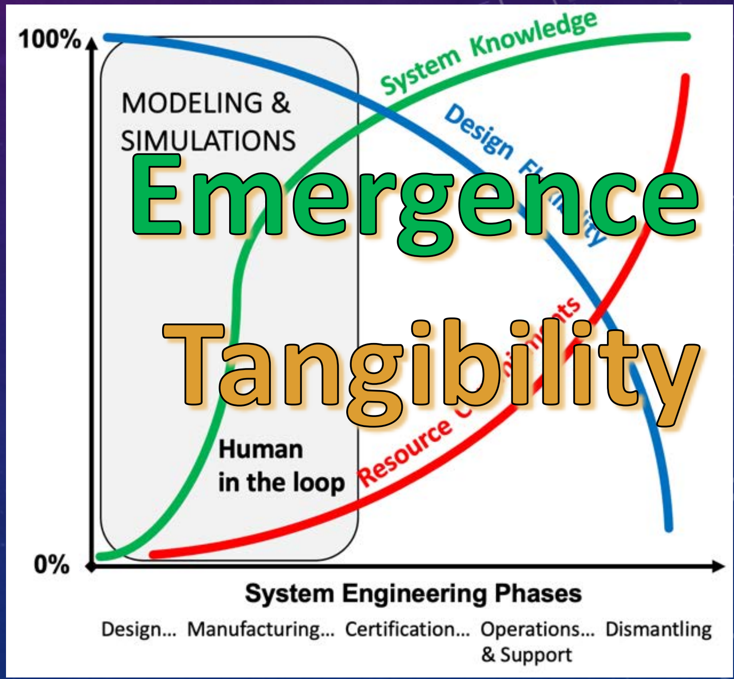
Organization (ORL)

ORL-0	**First principles where potential organizational models are explored.**
ORL-1	Goal-oriented research that requires making choices from first principles to practical fully digital organizational setups
ORL-2	Proof of principle development, and active R&D is started in a virtual environment
ORL-3	Virtual agile organizational prototype development and first HITLS (virtual HCD)
ORL-4	Proof of organizational concept development using concrete scenario-based design from fully virtual to more tangible environments
ORL-5	Assessing organization capability in terms of authority sharing (responsibility, accountability and control), trust, collaboration and coordination, for example
ORL-6	Real-world use-case tests in a wider variety of situations - tangibilization continues
ORL-7	Practical integration with respect to criteria such as safety, efficiency and comfort, at various levels of granularity of the organization – tangibilization continues
ORL-8	Readiness for effective implementation on a real site (fully tangible) based on personnel feedback for deployment approval
ORL-9	Deployment involving both personnel and real machines

TECHNOLOGY-CENTERED ENGINEERING:
STILL LAGGING BEHIND
IN THE LIFE CYCLE



HUMAN-CENTERED DESIGN:
WHAT WE REALLY WANT



DIG~~IT~~AL TWINS
VIRTUAL

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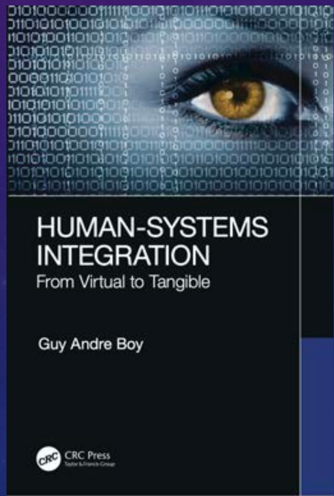
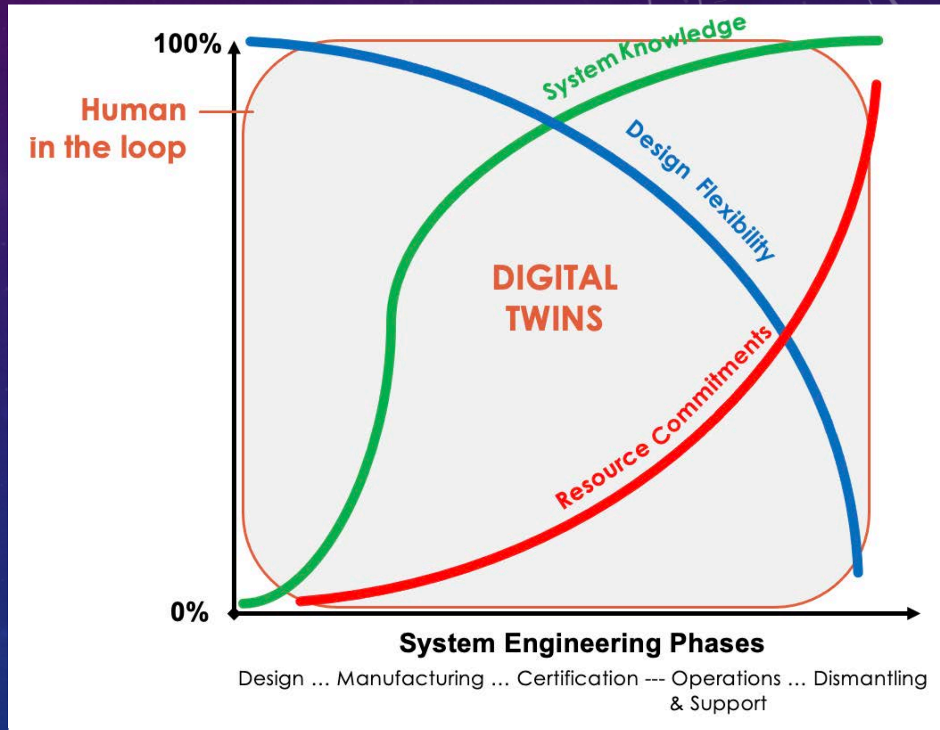
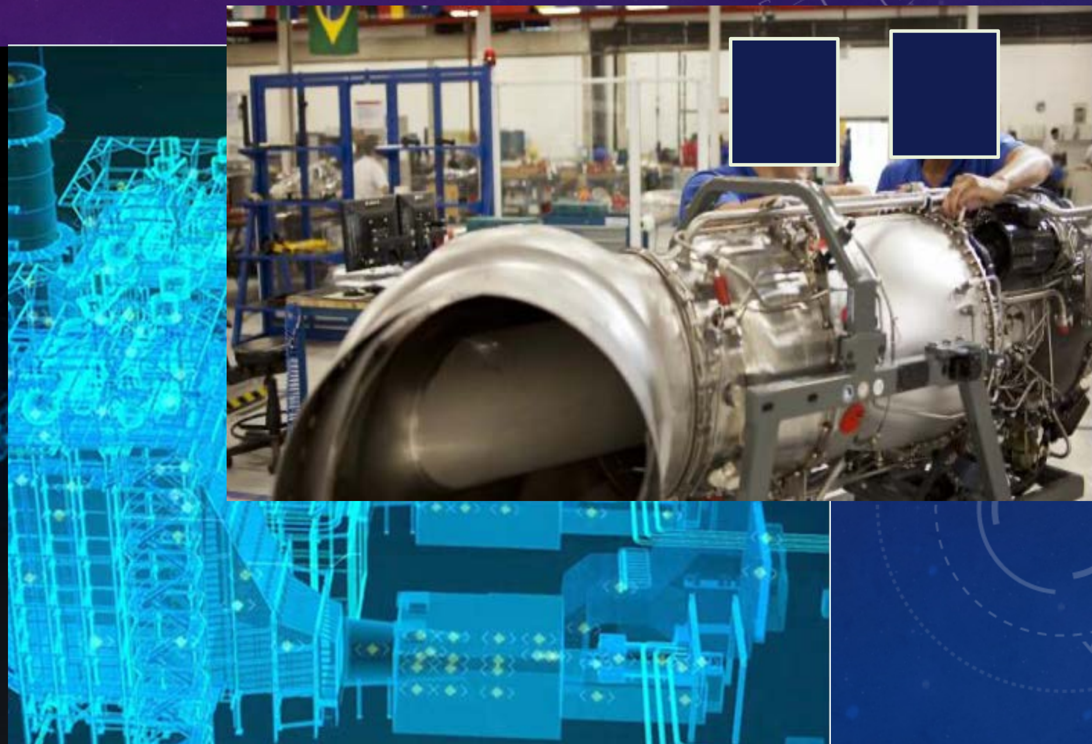
- Expanding human-in-the-loop simulations
    - during the whole life cycle of a system
    - "**what if?**"
- Vivid documentation
    - Experience feedback integration
    - **Organizational memory**
- Digital twins as Virtual Assistants
    - Multi-agent collaboration
    - Mediators for **collaborative work**





A DIGITAL TWIN FOR HELICOPTER ENGINE MAINTENANCE

Situational awareness, decision-making, trust & collaboration



CONCLUDING…

Awareness of the various possible contexts!
scenarios
human-in-the-loop simulations
elicitation of emergent cognitive functions

Scenario-based design → solid conceptual models

Orchestra Model pour design, evaluation and operations
Music theory → common framework (interaction models)
Composers → scores = contracts + coordination
Conductors → dynamic re-allocation
Musicians → **competence + engagement + cooperation**
Audience → constant communication and education

A FEW TAKE-AWAYS

- We live in a digital world → **tangibility** is a crucial contemporary issue
- Single-agent ergonomics is not enough → **Socio-ergonomics**
- Human-machine teaming → what **new human roles**?
- Rigid automation is what we know → **Flexible autonomy** is what we need to make
- How do we deal with the unexpected? → **problem-solving support**
- From means to purpose (people adapt) → **From purpose to means** (machines adapt)
- Collaborative work requires **education**, **openness**, **empathy** and **enthusiasm**!

This book is a follow-up of previous contributions in Human-Centered Design and practice in the development of virtual prototypes that requires progressive operational tangibility toward Human-Systems Integration (HSI). The book discusses flexibility in design and operations, tangibility of software-intensive systems, virtual human-centered design, increasingly-autonomous complex systems, Human-Factors and Ergonomics of sociotechnical systems, and systems of systems integration.

This is an attempt to better formalize a systemic approach to HSI. Good HSI is a matter of maturity… it takes time to mature. It takes time for a human being to become autonomous, and then mature! HSI is a matter of human-machine teaming, where human-machine cooperation and coordination are crucial. We cannot think engineering design without considering people and organizations that go with it. We also cannot think new technology, new organizations and new jobs without considering change management, especially in digital organizations.

The book will be of interest to industry, academia, those involved with systems engineering, human factors and the broader public.

Features:

- Discusses flexibility in design and operations of complex systems
- Offers tangibility of software-intensive systems
- Presents virtual human-centered design
- Covers autonomous complex systems
- Provides human factors and ergonomics of sociotechnical systems

About the Author:

Guy André Boy is one of the pioneers and a world leader in the study and applications of human centered design and human systems integration. He is also the Chair of INCOSE Human Systems Integration Working Group worldwide.

Ergonomics and Human Factors

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HUMAN-SYSTEMS INTEGRATION

Guy Andre Boy

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HUMAN-SYSTEMS INTEGRATION

From Virtual to Tangible

Guy Andre Boy

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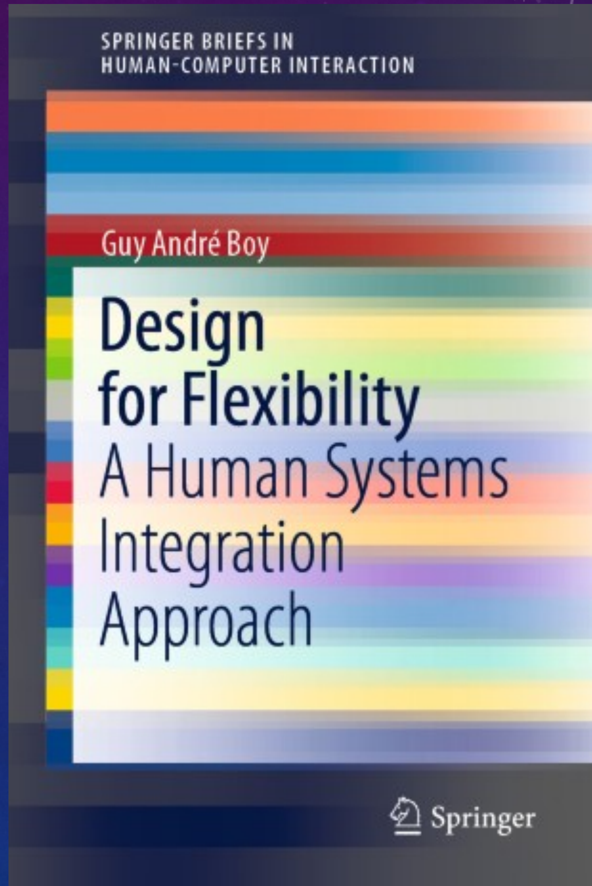
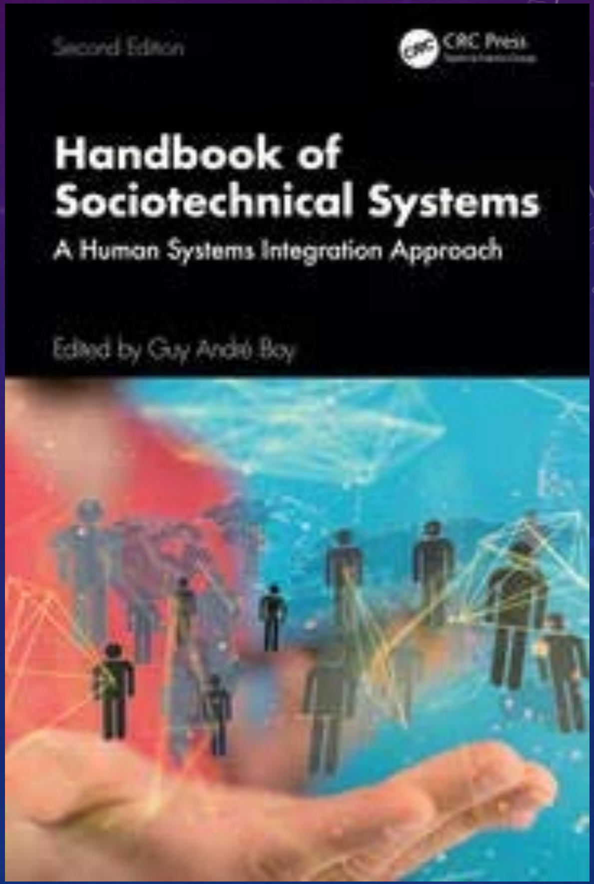
… another one on flexibility!



… and the last one!



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Thank you!